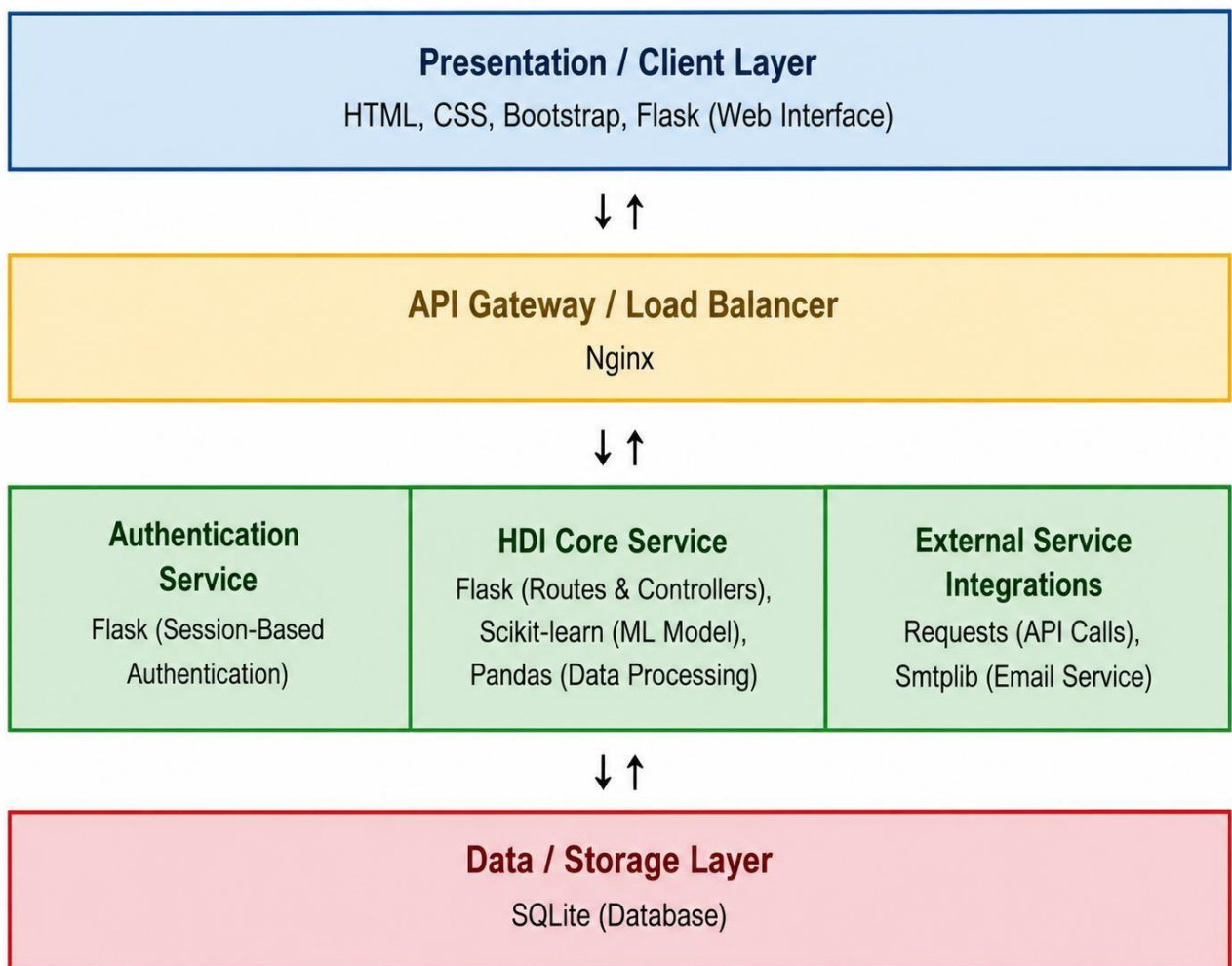


Solution Architecture

Date	18 June 2026
Team ID	
Project Name	A Comprehensive Measure of Well-Being
Maximum Marks	5 Marks

Solution Architecture Diagram:

Provide a high-level visual representation of your solution's architecture. Use the structural layout below as a template to map out your Presentation Layer, Application Layer, and Data Layer. Fill in the specific technologies and components for your project directly into the boxes to create your architectural diagram.



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Provides the user interface where users enter HDI indicators, view prediction results, charts, recommendations, and downloadable reports.	HTML, CSS, Bootstrap, JavaScript, Flask (Jinja2 Templates)
API Gateway / Application Layer	Receives user requests, validates input data, routes requests to the prediction module, and returns prediction results to the frontend.	Flask, Python
Core Service (ML Prediction Service)	Processes input data, performs preprocessing, loads the trained machine learning model, predicts the HDI score, classifies the HDI category, generates explanations, and provides recommendations.	Python, Scikit-learn, Pandas, NumPy, Joblib
Database / Storage Layer	Stores user accounts, prediction history, generated reports, and the trained machine learning model and dataset required for prediction.	SQLite (or MySQL), CSV Dataset, Joblib (.pkl Models)